

The Prolog Universe (PU) Manifesto

A Discrete Logic Inference Framework for Unified Physical Phenomena

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ABSTRACT

This paper proposes a departure from continuous, force-based physics in favor of a discrete, stateless logic engine model. By defining the universe as a recursive inference system, we demonstrate that General Relativity and Quantum Mechanics are not conflicting laws, but rather emergent behaviors of a system governed by finite computational bandwidth and recursive backtracking.

I. FUNDAMENTAL POSTULATES

- **Information Primacy:** The fundamental substrate of the universe is not matter or energy, but discrete logical units.
- **Discrete Execution:** Reality operates at a fixed maximum resolution (Planck length) and a fixed clock rate (Planck time).
- **Stateless Inference:** The universe is a live execution stream where current states are resolved on-demand through logical consistency, rather than being stored as a persistent historical state.

II. GRAVITATION AS COMPUTATIONAL LATENCY

In the PU model, gravity is redefined as a manifestation of **Processing Congestion**.

- **The Latency Gradient:** High-density coordinates (Mass) require more computational cycles to resolve. This creates localized update-frequency drops—effectively "Lag."

- **Dependency Propagation:** Every coordinate depends on the state of its neighbors to resolve its own next state. This creates a cascading wait-state that dilutes geometrically ($1/r^2$) through the 3D grid.
- **Trajectory Bias:** Particles do not "fall" due to attraction; they are statistically biased toward high-latency zones because the logical resolution of their next position is more stable in slower-updating regions.

III. QUANTUM PHENOMENA AS RECURSIVE BACKTRACKING

The "spookiness" of quantum events is resolved by the engine's inherent mechanism for maintaining logical integrity.

- **Superposition as Unresolved Variables:** A particle in flight is an unbound variable exploring multiple logical paths via a Redo loop. The interference pattern is the aggregate of all successful resolutions.
- **The Singleton Constraint (Measurement):** An observation introduces a "Ground Truth" fact. The engine performs a **Recursive Backtracking (Redo)** to satisfy this new constraint.
- **Retroactive Consistency:** The "Delayed Choice" paradox is a trivial byproduct of the engine's requirement for consistency. It is not "time travel"; it is the engine re-evaluating an active query to accommodate newly introduced facts.

IV. RELATIVISTIC EFFECTS AS RESOURCE ALLOCATION

Special Relativity is refactored as a **Bandwidth Limitation**.

- **Cycle Stealing:** Each coordinate has a fixed budget of processing cycles. High-velocity movement consumes a portion of these cycles for "Coordinate Translation."
- **Time Dilation:** Fewer cycles remain available for "Internal State Resolution" (aging/clocks). Time slows because the system prioritizes movement logic over internal processing.

- **The Universal Limit (c):** Represents the point at which 100% of available bandwidth is consumed by translation.

V. CONCLUSION

The shift from a Force-Based model to a Constraint-Based model eliminates the need for "Dark Matter" (unaccounted latency) and "Spooky Action" (logical re-evaluation). The universe is a masterpiece of **System Optimization**, satisfying a logic tree under the constraint of finite processing power.

Statement of Contribution:

This conceptual framework was developed by **Jan Hercek**, who provided the core logical architecture, system constraints, and theoretical synthesis. The formal technical drafting, structural organization, and terminological refinement were assisted by the **Gemini AI (Google)**, acting as a collaborative technical advisor. This partnership represents a human-centric model of theory development, where human intuition and "first-principles" logic are synthesized through high-scale computational modeling.